

B0° 21B5↵

```
clc;
clear all;
close all;
%x1x2x3
%
A = [3,2,1;2,3,1;1,2,3];
b = [39,34,26];
b = b';
```

```
fprintf(":")
```

:

```
if(A ~= 0 )
    X=(A\b)'
else%,
    fprintf("A is singular matrix,can't interation")
end
```

```
X = 1x3
    9.2500    4.2500    2.7500
```

```
ab = [A,b];%ab
N = 100;%
Epsilon =1e-4;
[R,C] = size(A);
ab_D = A;
ab_L = A;
for i=1:R
    for j=1:R
        if(i~=j)
            ab_D(i,j) = 0;
        end
        if(i>j)
            ab_L(i,j) = -ab_L(i,j);
        else
            ab_L(i,j) = 0;
        end
    end
end
end
```

```
ab_U = ab_D - ab_L - A;
```

1

```
T_J = ab_D^-1*(ab_L + ab_U);
C_J = ab_D^-1*b;
```

```
T_G = (ab_D - ab_L)^-1 * ab_U;
C_G = (ab_D-ab_L)^-1*b;
```

```
X_G = [0;0;0];
```

```
%
%JacobiGauss
if(Split_G(A)<1)
    i = 1;
    while(error_G>=Epsilon)
        X_G(:,i+1) = T_G*X_G(:,i) + C_G;
        error_G(i) = sqrt(sum((X_G(:,i+1)-X(1,:)).^2,1));
        i = i+1;
    end
    X_G = X_G';
    error_G
    X_G
else
    fprintf("Jacobi method can't convergent")
end
```

error_G = 1x15

4.0752	1.3142	0.5335	0.2393	0.1112	0.0523	0.0247	0.0117	0.005
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X_G = 16x3

0	0	0
13.0000	2.6667	2.5556
10.3704	3.5679	2.8313
9.6776	3.9378	2.8156
9.4363	4.1040	2.7853
9.3356	4.1812	2.7673
9.2901	4.2175	2.7583
9.2689	4.2346	2.7539
9.2589	4.2427	2.7519
9.2542	4.2466	2.7509

```
if(Split_J(A)<1)
    i = 1;
    while(error_J>=Epsilon)
        X_J(:,i+1) = T_J*X_J(:,i) + C_J;
        error_J(i) = sqrt(sum((X_J(:,i+1)-X(1,:)).^2,1));
        i = i+1;
    end
    X_J = X_J';
    error_J
    X_J
else
    fprintf("Jacobi method can't convergent")
end
```

```
end
```

```
Jacobi method can't convergent
```

```
function split = Split_J(A) %|TJ|
[R,C] = size(A);
ab_D = A;
for i=1:R
    for j=1:R
        if(i~=j)
            ab_D(i,j) = 0;
        end
    end
end

split = abs(eig(ab_D^-1 * (ab_D-A)));
split = max(split);

end
```

```
function split = Split_G(A) %|TG|
[R,C] = size(A);
ab_D = A;
ab_L = A;
for i=1:R
    for j=1:R
        if(i~=j)
            ab_D(i,j) = 0;
        end
        if(i>j)
            ab_L(i,j) = -ab_L(i,j);
        else
            ab_L(i,j) = 0;
        end
    end
end

ab_U = ab_D - ab_L - A;
```

```
split = abs(eig((ab_D-ab_L)^-1 * ab_U));  
split = max(split);  
  
end
```